

Ch. 3 – Point-to-Point Protocol (PPP)



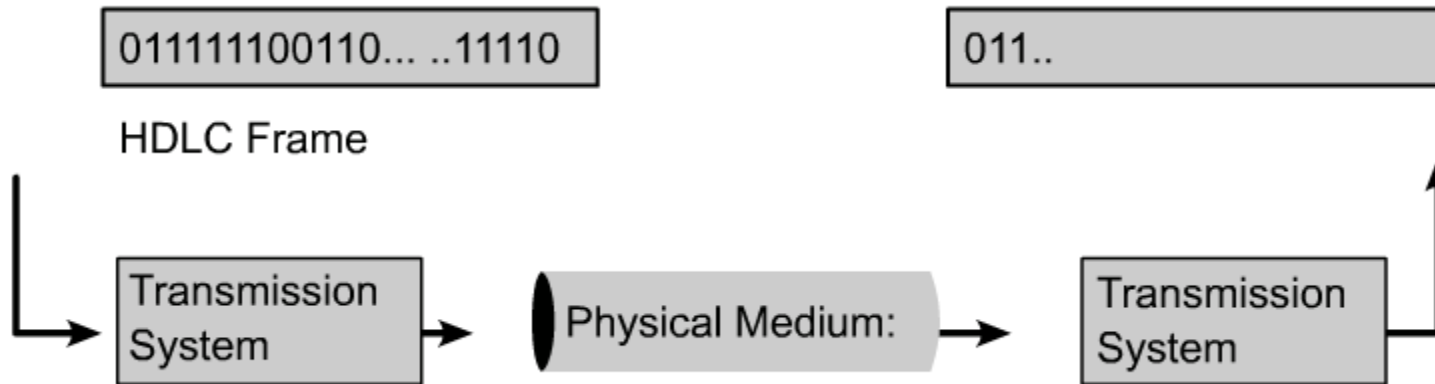
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CCNA 4 version 3.0

Overview

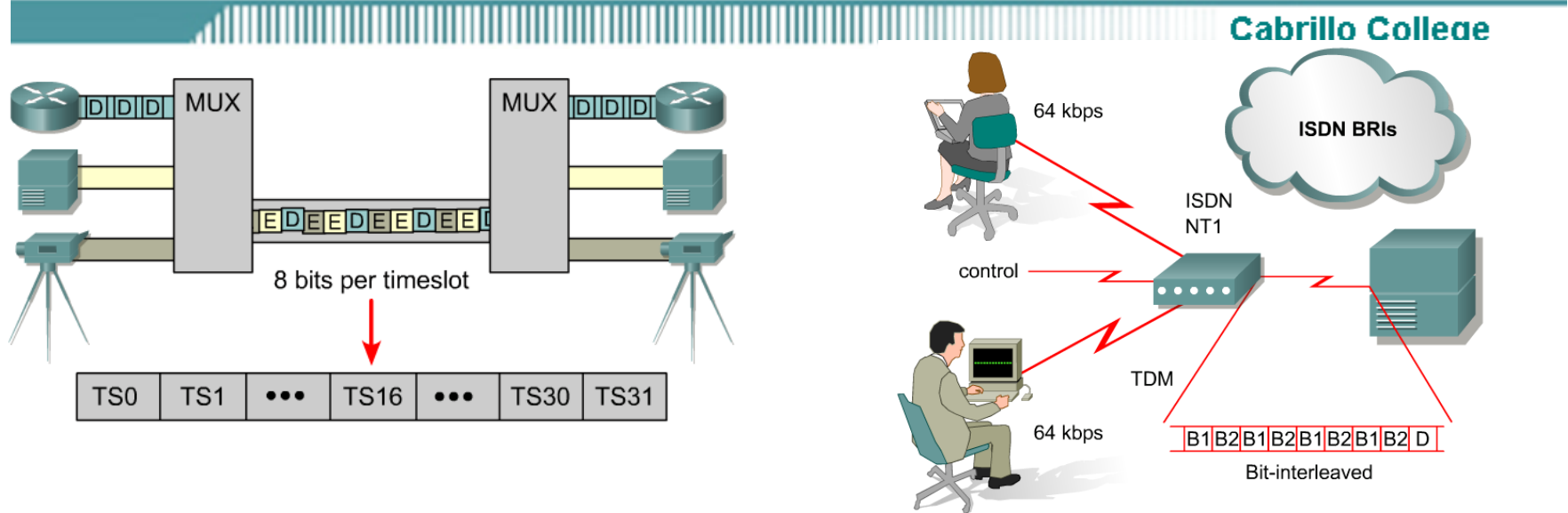
- Explain serial communication
- Describe and give an example of TDM
- Identify the demarcation point in a WAN
- Describe the functions of the DTE and DCE
- Discuss the development of HDLC encapsulation
- Use the **encapsulation hdlc** command to configure HDLC
- Troubleshoot a serial interface using the **show interface** and **show controllers** commands
- Identify the advantages of using PPP
- Explain the functions of the Link Control Protocol (LCP) and the Network Control Protocol (NCP) components of PPP
- Describe the parts of a PPP frame
- Identify the three phases of a PPP session
- Explain the difference between PAP and CHAP
- List the steps in the PPP authentication process
- Identify the various PPP configuration options
- Configure PPP encapsulation
- Configure CHAP and PAP authentication
- Use **show interface** to verify the serial encapsulation
- Troubleshoot any problems with the PPP configuration using **debug PPP**

Serial Communications



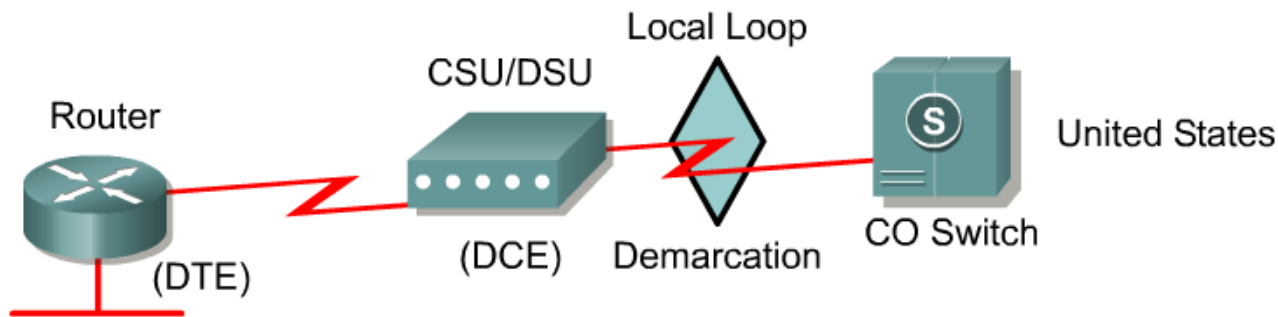
- WAN technologies are based on serial transmission at the physical layer.
- This means that the bits of a frame are transmitted one at a time over the physical medium.
- Some of the many different serial communications standards are the following:
 - RS-232-E
 - V.35
 - High Speed Serial Interface (HSSI)

Time Division Multiplexing



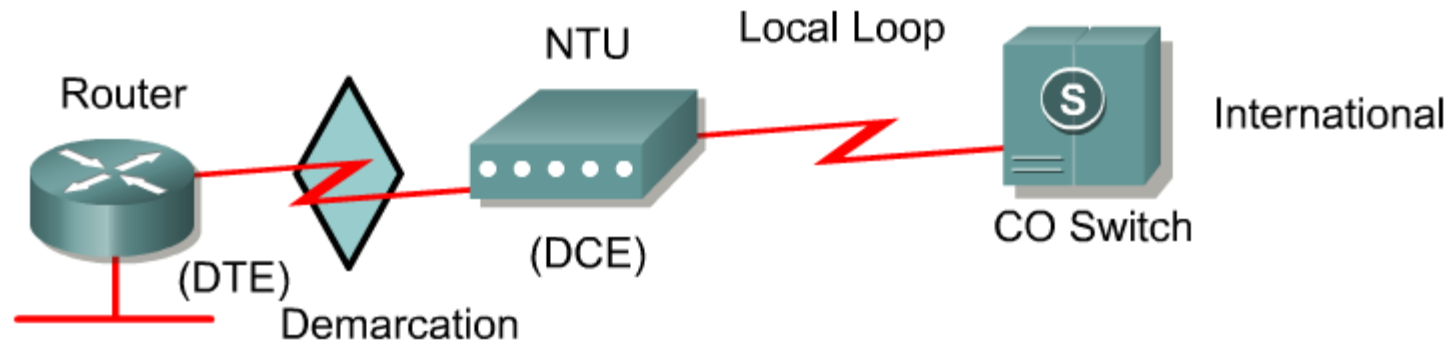
- Time-Division Multiplexing (TDM) is the transmission of several sources of information using one common channel, or signal, and then the reconstruction of the original streams at the remote end.
- In TDM, the output timeslot is always present whether or not the TDM input has any information to transmit.
- One TDM example is Integrated Services Digital Network (ISDN). ISDN basic rate (BRI) has three channels consisting of two 64 kbps B-channels (B1 and B2), and a 16 kbps D-channel.
- The TDM has nine timeslots, which are repeated.

Demarcation Point – U.S.



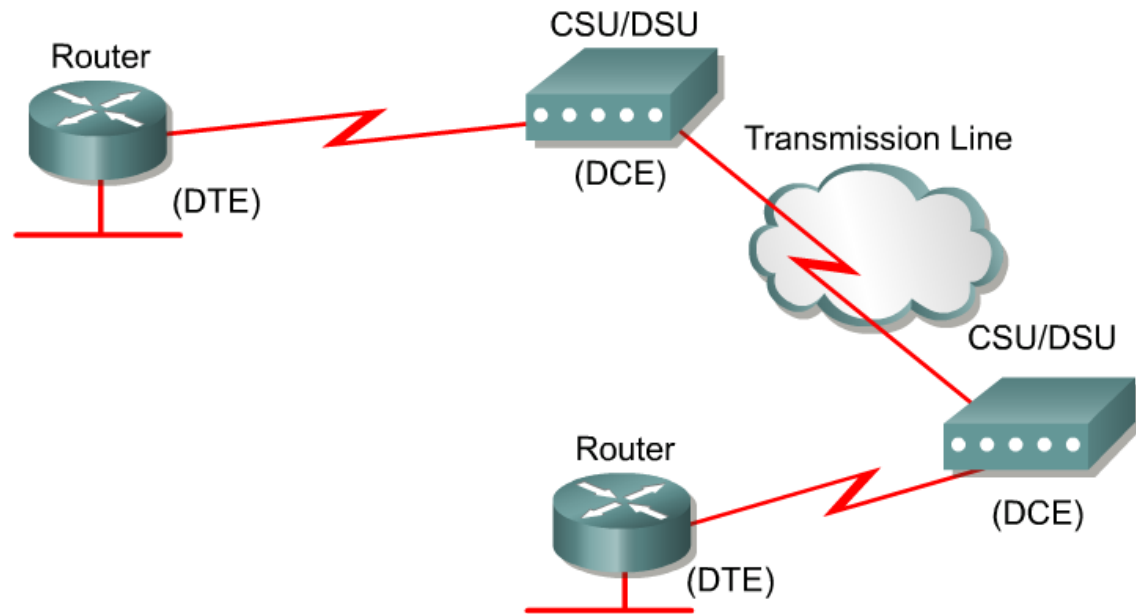
- The **demarcation point**, or "**demarc**" as it is commonly known, is the point in the network where the responsibility of the service provider or "telco" ends.
- In the United States, a telco provides the local loop into the customer premises and the customer provides the active equipment such as the channel service unit/data service unit (CSU/DSU) on which the local loop is terminated.
- This termination often occurs in a telecommunications closet and the customer is responsible for maintaining, replacing, or repairing the equipment.

Demarcation Point – International



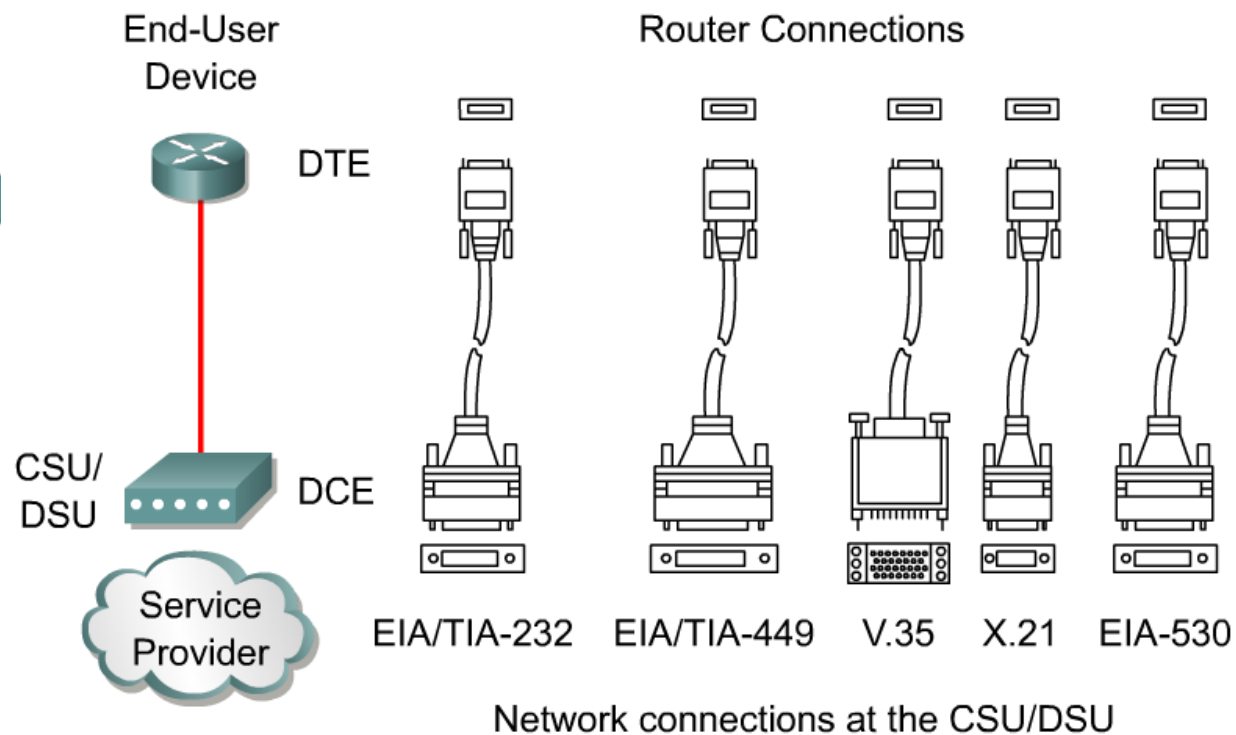
- In other countries around the world, the **network terminating unit (NTU)** is provided and managed by the telco.
- This allows the telco to actively manage and troubleshoot the local loop with the demarcation point occurring after the NTU.
- The customer connects a customer premises equipment (CPE) device, such as a router or frame relay access device, into the NTU using a V.35 or RS-232 serial interface.

DTE-DCE



- Many standards have been developed to allow DTEs to communicate with DCEs.
- The Electronics Industry Association (EIA) and the International Telecommunication Union Telecommunications Standardization Sector (ITU-T) have been most active in the development of these standards.

DTE-DCE



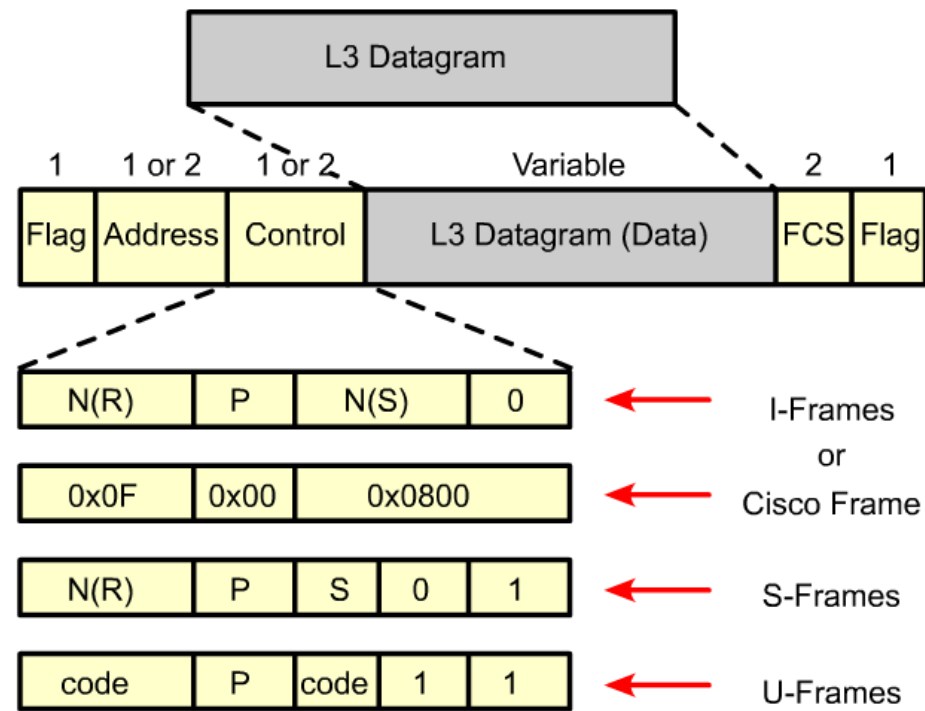
- The DTE-DCE interface for a particular standard defines the following specifications:
- **Mechanical/physical** – Number of pins and connector type
- **Electrical** – Defines voltage levels for 0 and 1
- **Functional** – Specifies the functions that are performed by assigning meanings to each of the signaling lines in the interface
- **Procedural** – Specifies the sequence of events for transmitting data

DTE-DCE



- If two DTEs must be connected together, like two computers or two routers in the lab, a special cable called a null-modem is necessary to eliminate the need for a DCE.
- For synchronous connections, where a clock signal is needed, either an external device or one of the DTEs must generate the clock signal.
- To support higher densities in a smaller form factor, Cisco has introduced a smart serial cable.
- The serial end of the smart serial cable is a 26-pin connector significantly more compact than the DB-60 connector.

HDLC Encapsulation



- In 1979, the ISO agreed on HDLC as a standard bit-oriented data link layer protocol that encapsulates data on synchronous serial data links.
- Since 1981, ITU-T has developed a series of HDLC derivative protocols.
- The following examples of derivative protocols are called link access protocols:
 - Link Access Procedure, Balanced (LAPB) for X.25
 - Link Access Procedure on the D channel (LAPD) for ISDN
 - Link Access Procedure for Modems (LAPM) and PPP for modems
 - Link Access Procedure for Frame Relay (LAPF) for Frame Relay

HDLC Encapsulation

HDLC ISO frame

Flag	Address	Control	Data (Payload)	FCS	Flag
1 byte	1 byte	1 or 2 bytes	1500 bytes	2 (or 4) bytes	1 byte

PPP frame

Flag	Address	Control	Protocol	Data (Payload)	FCS	Flag
1 byte	1 byte	1 byte	1 or 2 bytes	Up to 1500 bytes	2 (or 4) bytes	1 byte

- Standard HDLC does not inherently support multiple protocols on a single link, as it does not have a way to indicate which protocol is being carried.
- Cisco offers a proprietary version of HDLC.
- The Cisco HDLC frame uses a proprietary 'type' field that acts as a protocol field.
- HDLC is the default Layer 2 protocol for Cisco router serial interfaces.
- PPP actually uses HDLC as a basis for encapsulating datagrams.

Configuring HDLC

```
Router(config-if)#encapsulation hdlc
```

- The default encapsulation method used by Cisco devices on synchronous serial lines is Cisco HDLC.
- Cisco HDLC is a point-to-point protocol that can be used on leased lines between two Cisco devices.
- When communicating with a non-Cisco device, synchronous PPP is a more viable option.

Troubleshooting a serial interface

```
Router#show interfaces s0/0
```

```
Serial 0 is up, line protocol is up
```

```
Hardware is MCI Serial
```

```
Internet address is 131.108.156.98, subnet mask is  
255.255.255.240
```

```
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec, rely  
255/255, load 1/255
```

```
Encapsulation HDLC, loopback not set, keepalive set  
(10 sec)
```

```
Router#show controllers serial 0/0
```

```
Interface Serial0/0
```

```
Hardware is PowerQUICC MPC860
```

```
DTE V.35 TX and RX clocks detected.
```

```
idb at 0x81414E2C, driver data structure at 0x8141753C
```

```
SCC Registers:
```

Most of these commands will not make sense until we discuss PPP and Frame Relay

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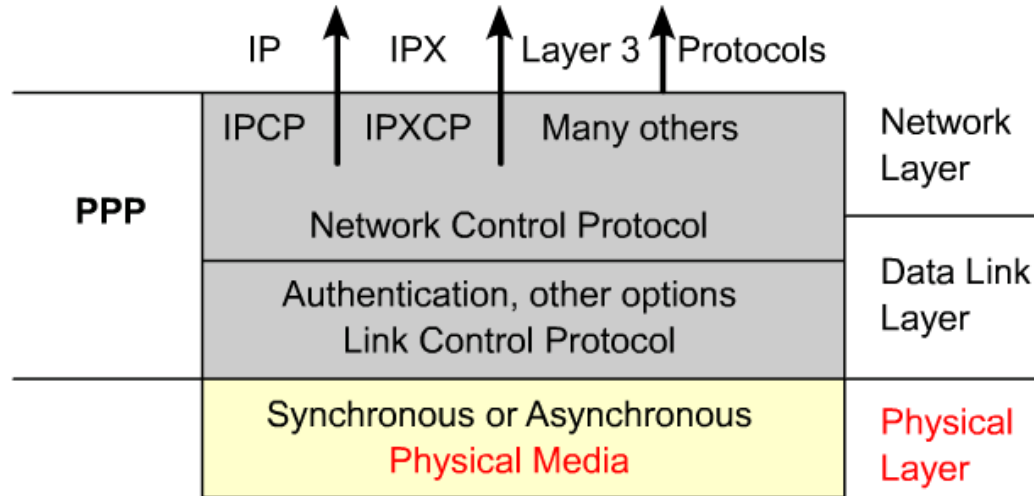
- **debug serial interface** – Verifies whether HDLC keepalive packets are incrementing. If they are not, a possible timing problem exists on the interface card or in the network.
- **debug arp** – Indicates whether the router is sending information about or learning about routers (with ARP packets) on the other side of the WAN cloud. Use this command when some nodes on a TCP/IP network are responding, but others are not.
- **debug frame-relay lmi** – Obtains Local Management Interface (LMI) information which is useful for determining whether a Frame Relay switch and a router are sending and receiving LMI packets.
- **debug frame-relay events** – Determines whether exchanges are occurring between a router and a Frame Relay switch.
- **debug ppp negotiation** – Shows Point-to-Point Protocol (PPP) packets transmitted during PPP startup where PPP options are negotiated.
- **debug ppp packet** – Shows PPP packets being sent and received. This command displays low-level packet dumps.
- **debug ppp** – Shows PPP errors, such as illegal or malformed frames, associated with PPP connection negotiation and operation.
- **debug ppp authentication** – Shows PPP Challenge Handshake Authentication Protocol (CHAP) and Password Authentication Protocol (PAP) packet exchanges.

PPP



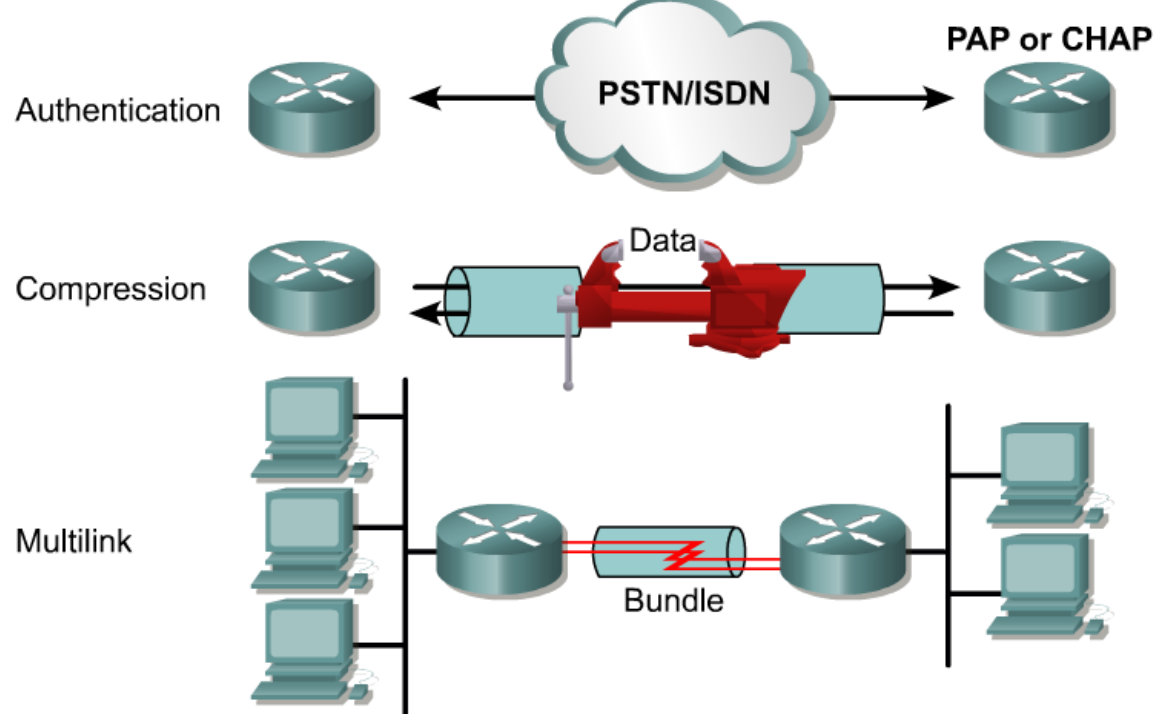
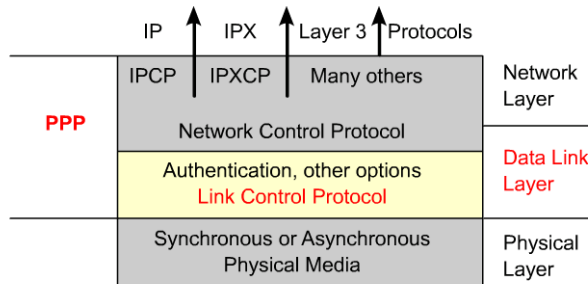
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PPP layered architecture



- PPP contains two sub-protocols:
 - **Link Control Protocol** – Used for establishing the point-to-point link.
 - Negotiate and setup control options on the WAN data link.
 - **Network Control Protocol** – Used for configuring the various network layer protocols.
 - Encapsulate and negotiate options for multiple network layer protocols.
 - The LCP sits on top of the physical layer and is used to establish, configure, and test the data-link connection.

LCP

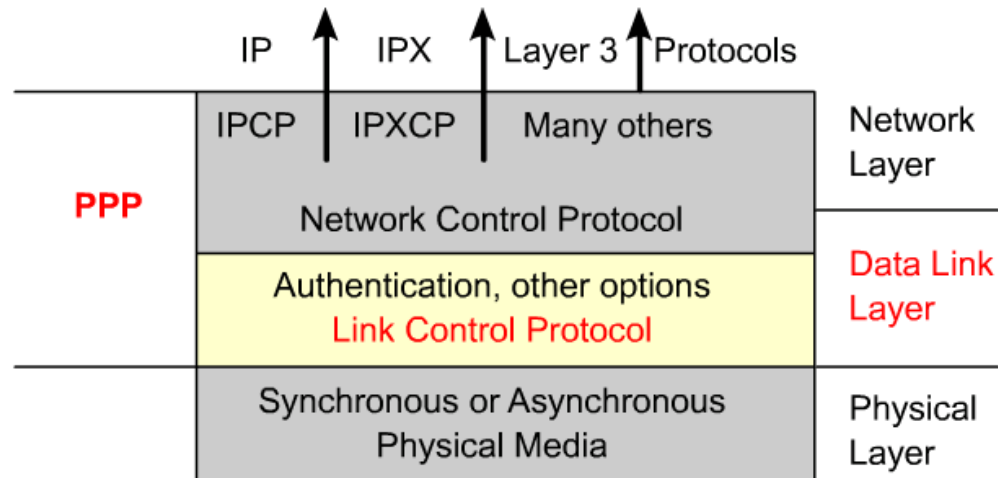


Features	How It Operates	Protocol
Authentication	Require a password and perform challenge handshake	PAP CHAP
Compression	Compress data at source and reproduce data at destination	Stacker, Predictor, TCP Header, or MPPC
Error Detection	Monitor data dropped on link Avoid frame looping	Quality Magic Number
Multilink	Load balancing across multiple links	Multilink Protocol (MP)

Also: PPP callback

- LCP is used to automatically agree upon encapsulation format options.

LCP



- LCP will also do the following:
 - Handle varying limits on packet size
 - Detect common misconfiguration errors
 - Terminate the link
 - Determine when a link is functioning properly or when it is failing

PPP Session Establishment

PPP Session Establishment

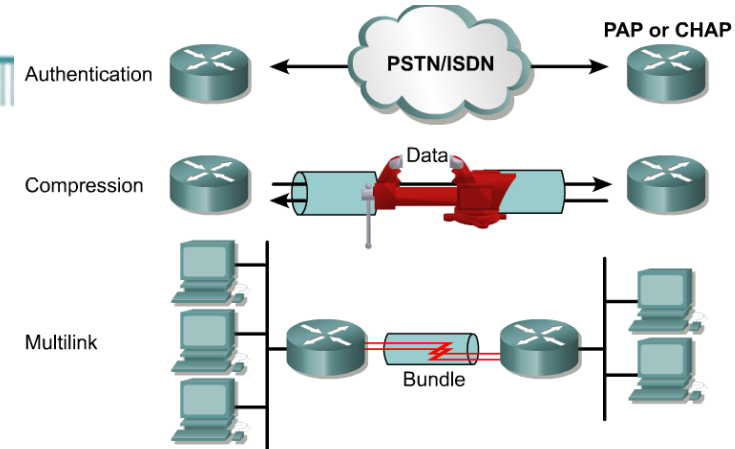
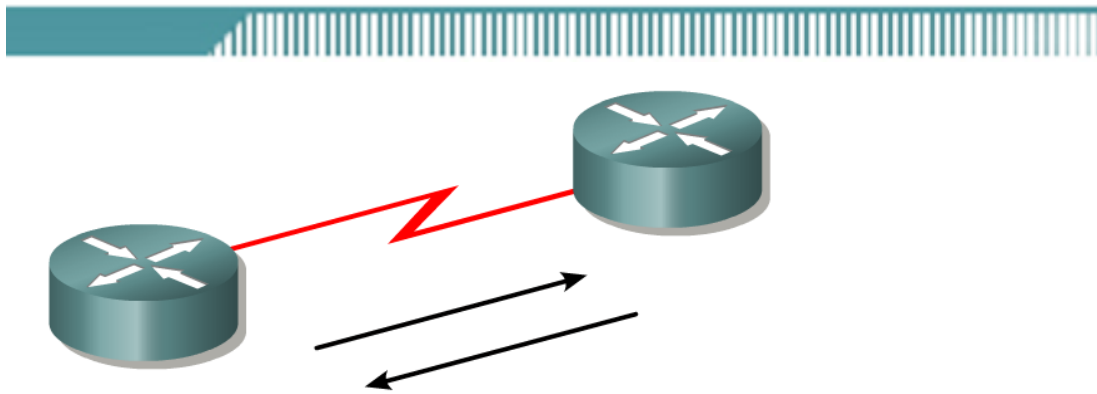
- Link Establishment Phase
- Optional Authentication Phase
- Network Layer Protocol Phase

- PPP session establishment progresses through three phases:
 - link establishment
 - authentication
 - network layer protocol phase

PPP Session Establishment (Detail)

1. Link establishment - (LCPs)
2. Authentication - Optional (LCPs)
3. Link quality determination - Optional (LCPs)
4. Network layer protocol configuration (NCPs)
5. Link termination (LCPs)

Link-establishment phase



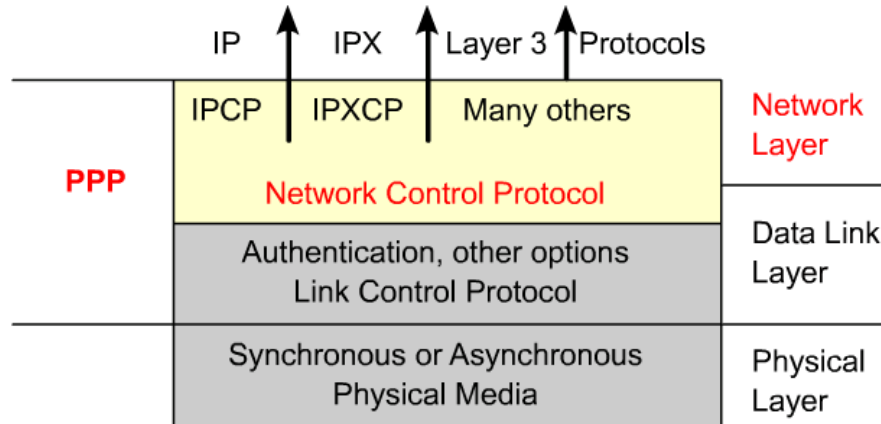
- In this phase each PPP device sends **LCP frames** to **configure and test the data link**.
- LCP frames contain a configuration option field that allows devices to negotiate the use of options such as the maximum transmission unit (**MTU**), **compression** of certain PPP fields, and the **link-authentication** protocol.
- If a configuration option is not included in an LCP packet, the default value for that configuration option is assumed.
- Before any network layer packets can be exchanged, LCP must first **open the connection and negotiate the configuration parameters**.
- This phase is complete when a configuration acknowledgment frame has been sent and received.

Authentication Phase (Optional)



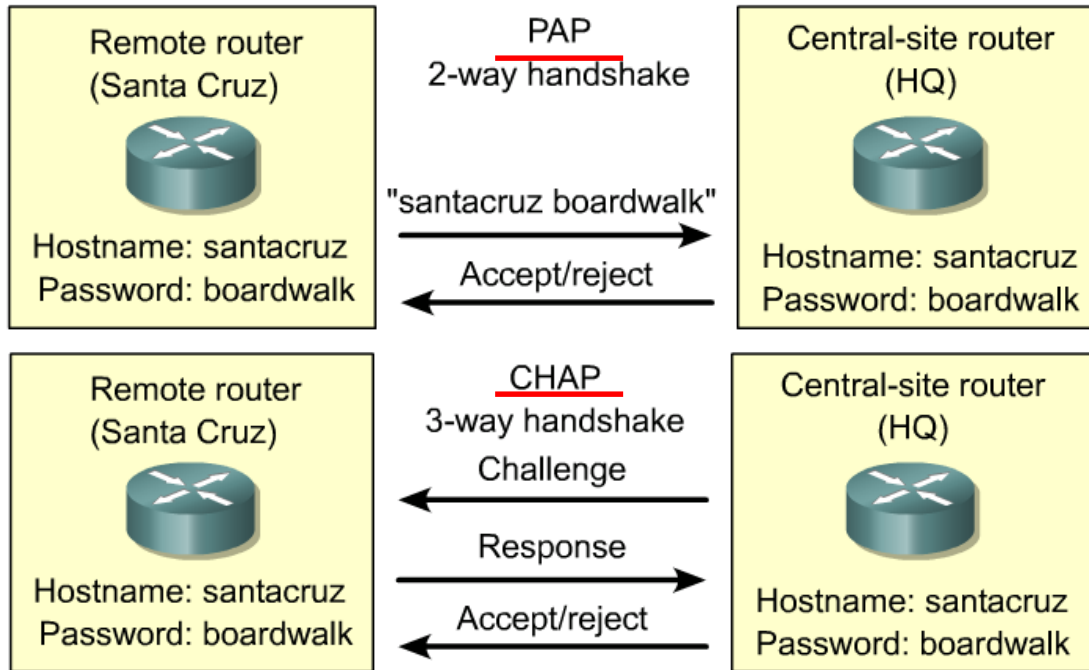
- After the link has been established and the **authentication protocol decided** on, the peer may be authenticated.
- Authentication, if used, takes place before the network layer protocol phase is entered.
- As part of this phase, LCP also allows for an **optional link-quality determination test**.
 - The link is tested to determine whether the **link quality is good enough** to bring up network layer protocols

Network Layer Protocol Phase



- In this phase the PPP devices send **NCP packets** to choose and **configure one or more network layer protocols**, such as IP.
- Once each of the chosen network layer protocols has been configured, packets from each network layer protocol can be sent over the link.
- If LCP closes the link, it informs the network layer protocols so that they can take appropriate action.
- The **show interfaces** command reveals the LCP and NCP states under PPP configuration.
- The PPP link remains configured for communications until LCP or NCP frames close the link or until an inactivity timer expires or a user intervenes.

PPP authentication protocols

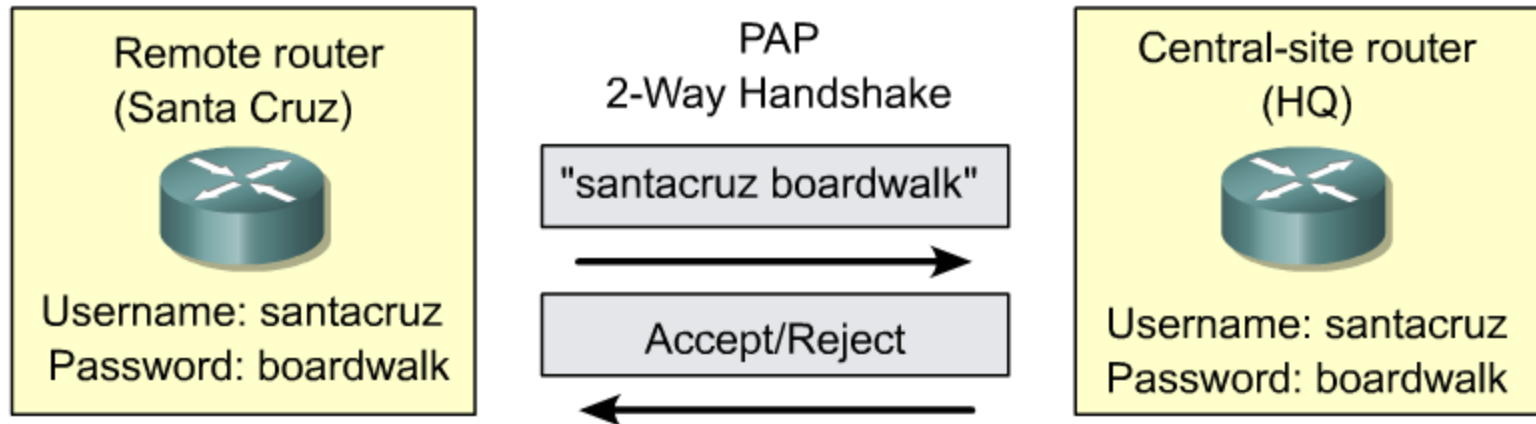


- Password sent in clear text
- Peer in control of attempts

Encrypted password
Repeated challenges

1. Link establishment - (LCPs)
- 2. Authentication - Optional (LCPs)**
3. Link quality determination - Optional (LCPs)
4. Network layer protocol configuration (NCPs)
5. Link termination (LCPs)

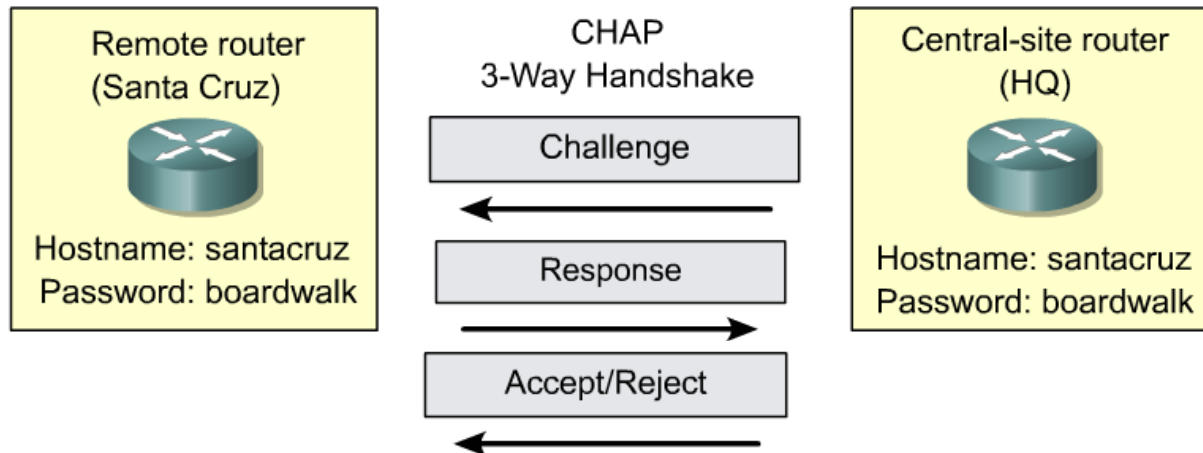
Password Authentication Protocol (PAP)



- PAP provides a simple method for a remote node to establish its identity, **using a two-way handshake**.
- After the PPP link establishment phase is complete, a **username/password pair is repeatedly sent** by the remote node across the link until authentication is acknowledged or the connection is terminated.
- PAP is not a strong authentication protocol.
- **Passwords are sent across the link in clear text** and there is no protection from playback or repeated trial-and-error attacks.
- The remote node is in control of the frequency and timing of the login attempts.

Challenge Handshake Authentication Protocol (CHAP)

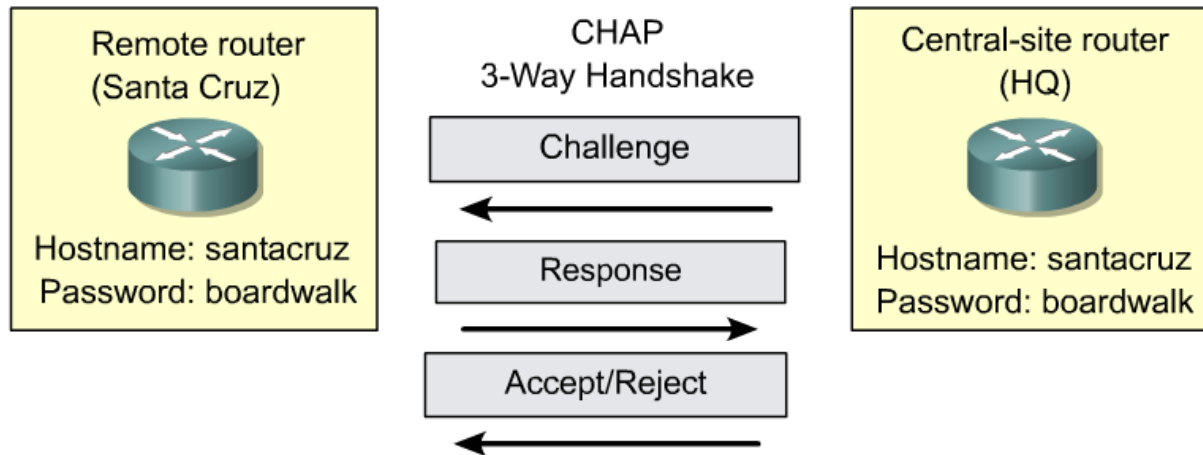
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- CHAP is used at the startup of a link and periodically verifies the identity of the remote node using a three-way handshake.
- After the PPP link establishment phase is complete, the local router sends a "challenge" message to the remote node.
- The remote node responds with a value calculated using a one-way hash function, which is typically Message Digest 5 (MD5).
- This response is based on the password and challenge message.
- The local router checks the response against its own calculation of the expected hash value.
- If the values match, the authentication is acknowledged, otherwise the connection is immediately terminated.

Challenge Handshake Authentication Protocol (CHAP)

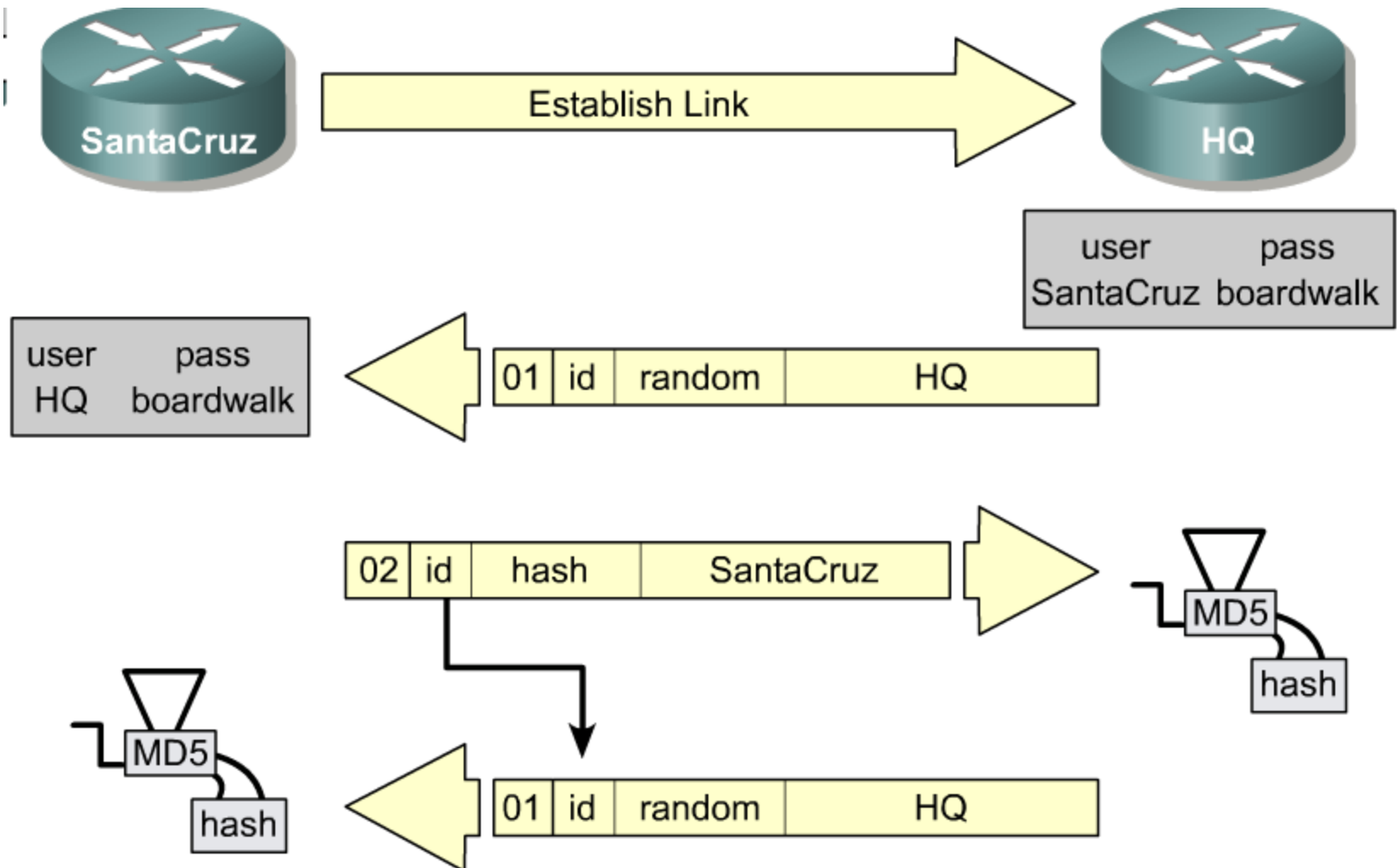
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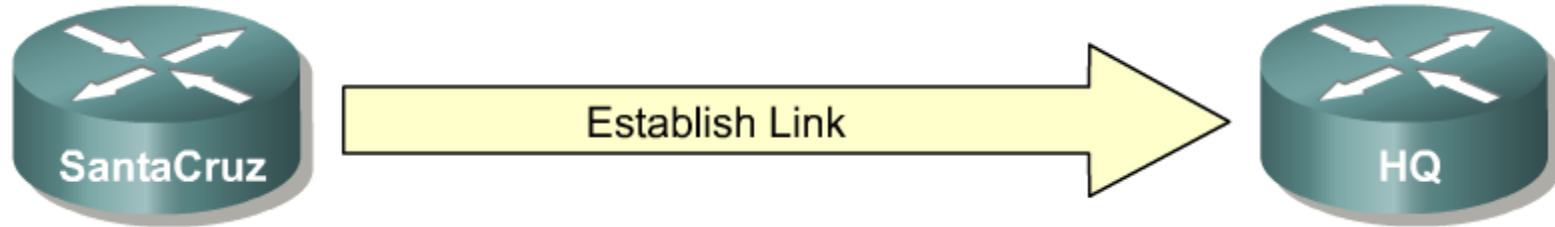
- CHAP provides protection against playback attack through the use of a variable challenge value that is unique and unpredictable.
- Since the challenge is unique and random, the resulting hash value will also be unique and random.
- The use of repeated challenges is intended to limit the time of exposure to any single attack.
- The local router or a third-party authentication server is in control of the frequency and timing of the challenges.

CHAP Operation

Note: A simpler version will be shown when we configure CHAP.

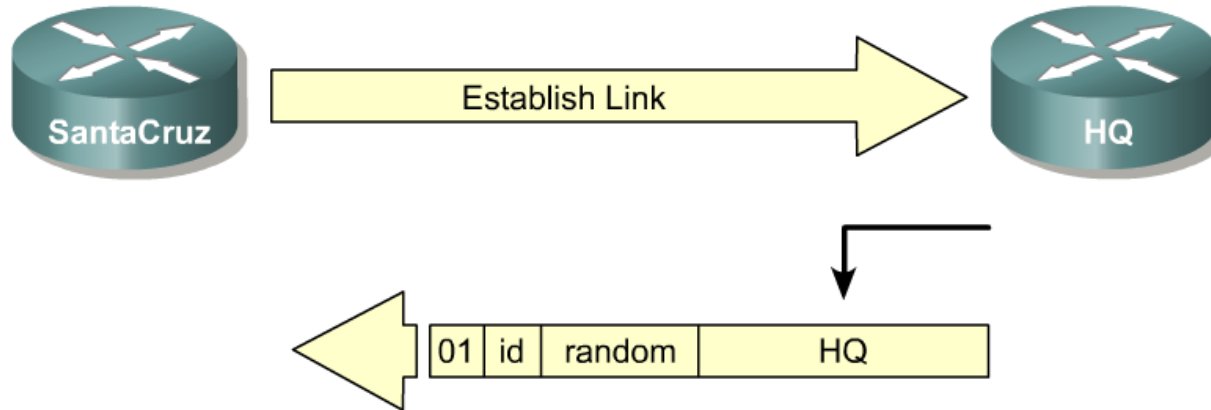


LCP establishes and negotiates the link



1. The call comes in to HQ. The incoming interface is configured with the **ppp authentication chap** command.
2. LCP negotiates CHAP and MD5.
3. A CHAP challenge from HQ to the calling router is required on this call.

CHAP Challenge

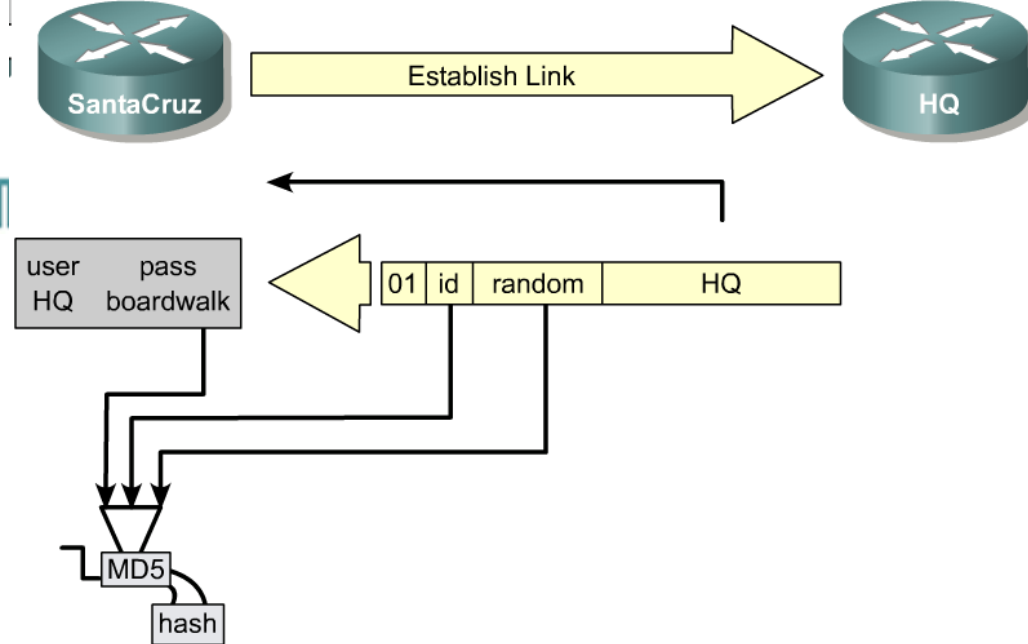


This figure illustrates the following steps in the CHAP authentication between the two routers:

1. A CHAP challenge packet is built with the following characteristics:
 - **01** = challenge packet type identifier.
 - **ID** = sequential number that identifies the challenge.
 - **random** = a reasonably random number generated by the router.
 - **HQ** = the authentication name of the challenger.
2. The **ID** and **random** values are kept on the called router.
3. The challenge packet is sent to the calling router. A list of outstanding challenges is maintained.

Receipt of the CHAP Challenge

- This diagram illustrates the receipt and MD5 processing of the challenge packet from the peer.
- The router processes the incoming CHAP challenge packet in the following manner:

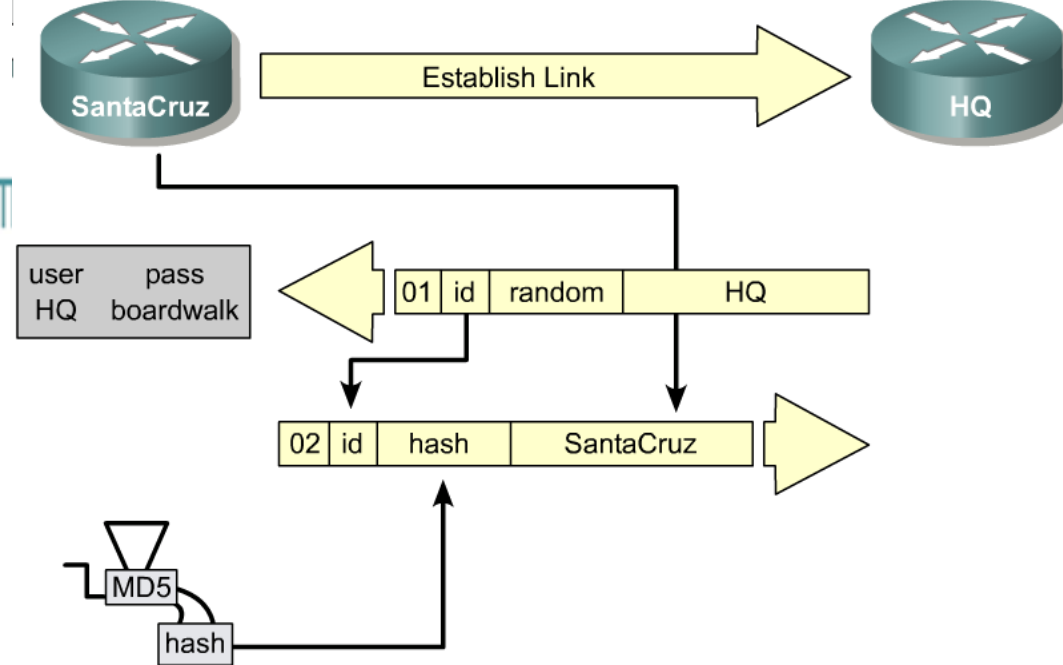


1. The **ID** value is fed into the MD5 hash generator.
2. The **random** value is fed into the MD5 hash generator.
3. The name **HQ** is used to look up the password. The router looks for an entry matching the username in the challenge. In this example, it looks for:

username HQ password boardwalk
4. The **password** is fed into the MD5 hash generator.
5. The result is the one-way **MD5-hashed** CHAP challenge that will be sent back in the CHAP response.

CHAP Response

- This diagram illustrates how the CHAP response packet sent to the authenticator is built.
- The following steps are shown in this figure:

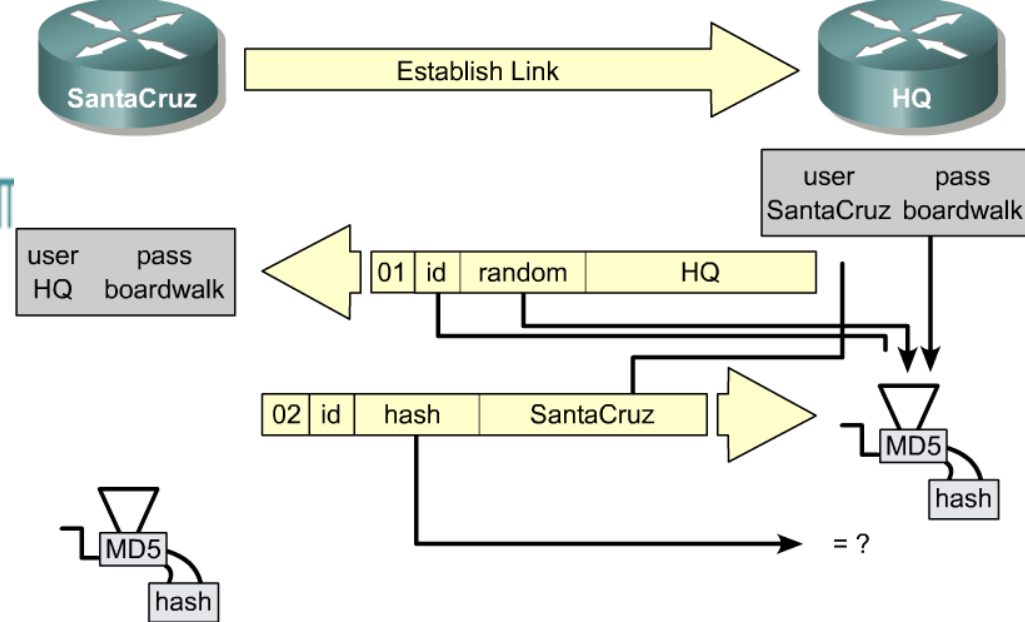


1. The response packet is assembled from the following components:
 - **02** = CHAP response packet type identifier.
 - **ID** = copied from the challenge packet.
 - **hash** = the output from the MD5 hash generator (the hashed information from the challenge packet).
 - **SantaCruz** = the authentication name of this device. This is needed for the peer to look up the username and password entry needed to verify identity (this is explained in more detail below).
2. The response packet is then sent to the challenger.

Receive CHAP Response

- This diagram shows how the challenger processes the response packet.
- The CHAP response packet is processed (on the authenticator) in the following manner:

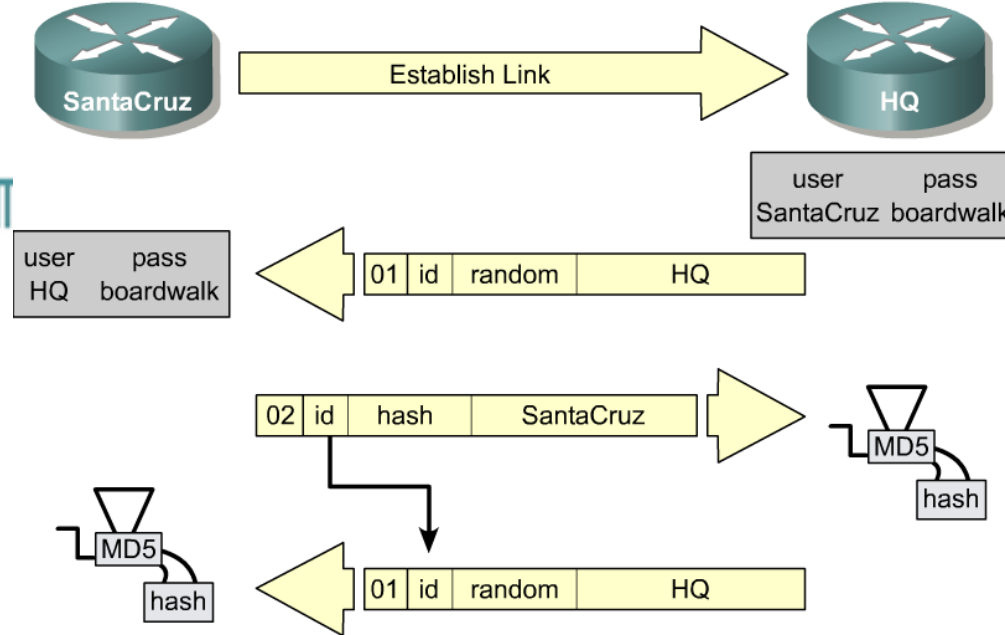
1. The **ID** is used to find the original challenge packet.
2. The **ID** is fed into the MD5 hash generator.
3. The original challenge **random** value is fed into the MD5 hash generator.
4. The name **SantaCruz** is used to look up the password from one of the following sources:
 - Local username and password database
 - **username SantaCruz password boardwalk**
 - RADIUS or TACACS+ server.
5. The **password** is fed into the MD5 hash generator.
6. The **hash** value received in the response packet is then **compared to the calculated MD5 hash value**. CHAP authentication succeeds if the calculated and the received hash values are equal.



Success

Message Sent

- This diagram illustrates the success message being sent to the calling router.



- If authentication is successful, a CHAP success packet is built from the following components:
 - **03** = CHAP success message type.
 - **ID** = copied from the response packet.
 - **“Welcome in”** is simply a text message providing a user-readable explanation.
- If authentication fails, a CHAP failure packet is built from the following components:
 - **04** = CHAP failure message type.
 - **ID** = copied from the response packet.
 - **“Authentication failure”** or other text message, providing a user-readable explanation.
- The success or failure packet is then sent to the calling router.

Configuring PPP

```
Router#configure terminal  
Router(config)#interface serial 0/0  
Router(config-if)#encapsulation ppp
```

- Enables PPP encapsulation on serial interface 0/0

Configuring PPP



DTE
.2/S0

172.25.3.0/24
Serial

DCE
.1/S0



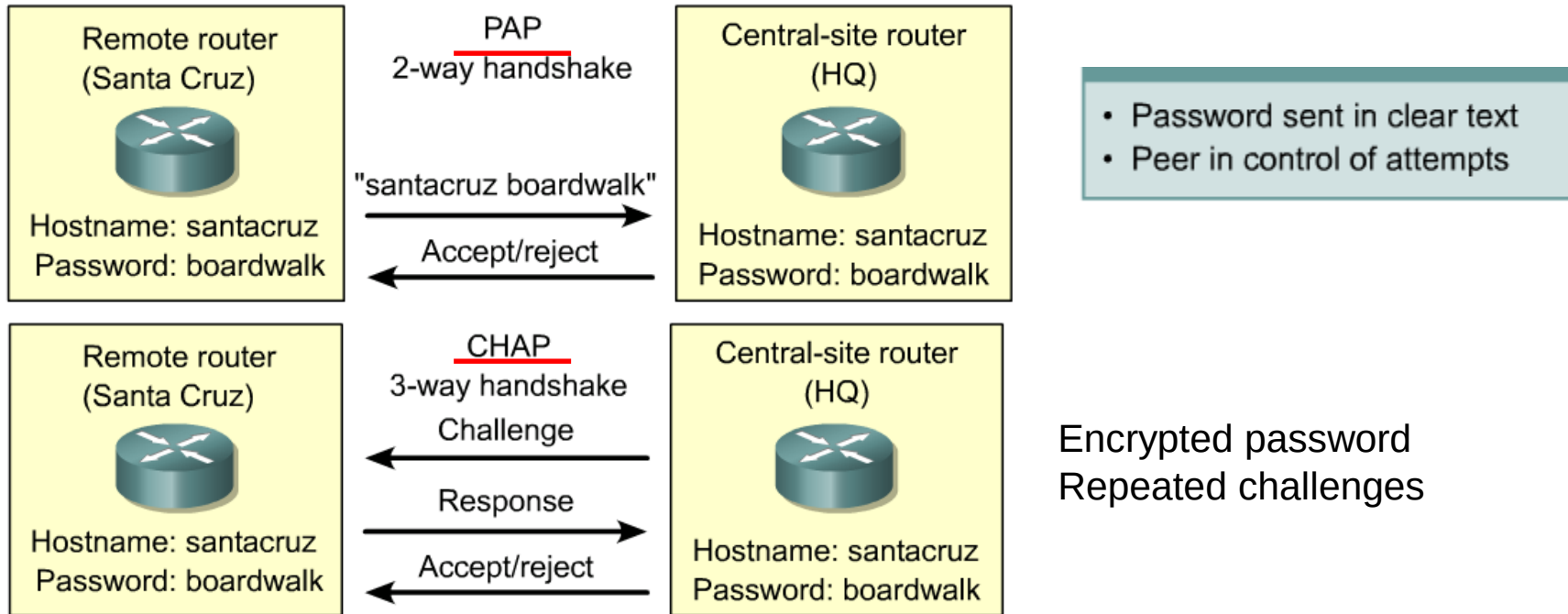
```
interface Serial0
  ip address 172.25.3.2 255.255.255.0
  encapsulation ppp
```

```
interface Serial0
  ip address 172.25.3.1 255.255.255.0
  encapsulation ppp
```

Verifying PPP

```
Router#show interfaces serial0/0
Serial0/0 is up, line protocol is up
  Hardware is HD64570
  Internet address is 10.140.1.2/24
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
  rely 255/255, load 1/255
  Encapsulation PPP, loopback not set, keepalive
  set (10 sec)
  LCP Open ← LCP
  Open: IPCP, CDPCP ← NCP
  Last input 00:00:05, output 00:00:05, output
  hang never
  Last clearing of "show interface" counters never
  Queueing strategy: fifo
  Output queue 0/40, 0 drops; input queue 0/75, 0
  drops
  5 minute input rate 0 bits/sec, 0 packets/sec
  -
```

Configuring Authentication (PAP or CHAP)



- Peer routers exchange authentication messages.
- Two alternatives are:
 - Password Authentication Protocol (PAP)
 - Challenge Handshake Authentication Protocol (CHAP)
- In general, CHAP is the preferred protocol but PAP is still very common.

Configuring PAP

```
Rtr(config) # username remote-host password remote-  
password
```

- This needs to match the ppp pap sent-username on the remote host.

```
Rtr(config-if) # ppp pap sent-username this-host  
username password this-host-password
```

- The passwords do not need to match between the remote and the host.
- It should not need to be the same as the enable-secret password.

```
Router(config-if) #ppp authentication {chap | chap pap  
| pap chap | pap}
```

- Two choices: first choice | second choice
- If both methods are enabled, then the first method specified will be requested during link negotiation.
- If the peer suggests using the second method or simply refuses the first method, then the second method will be tried.

Configuring PAP



DTE
.2/S0

172.25.3.0/24
Serial

DCE
.1/S0



```
hostname SantaCruz
username HQ password HQpass
```

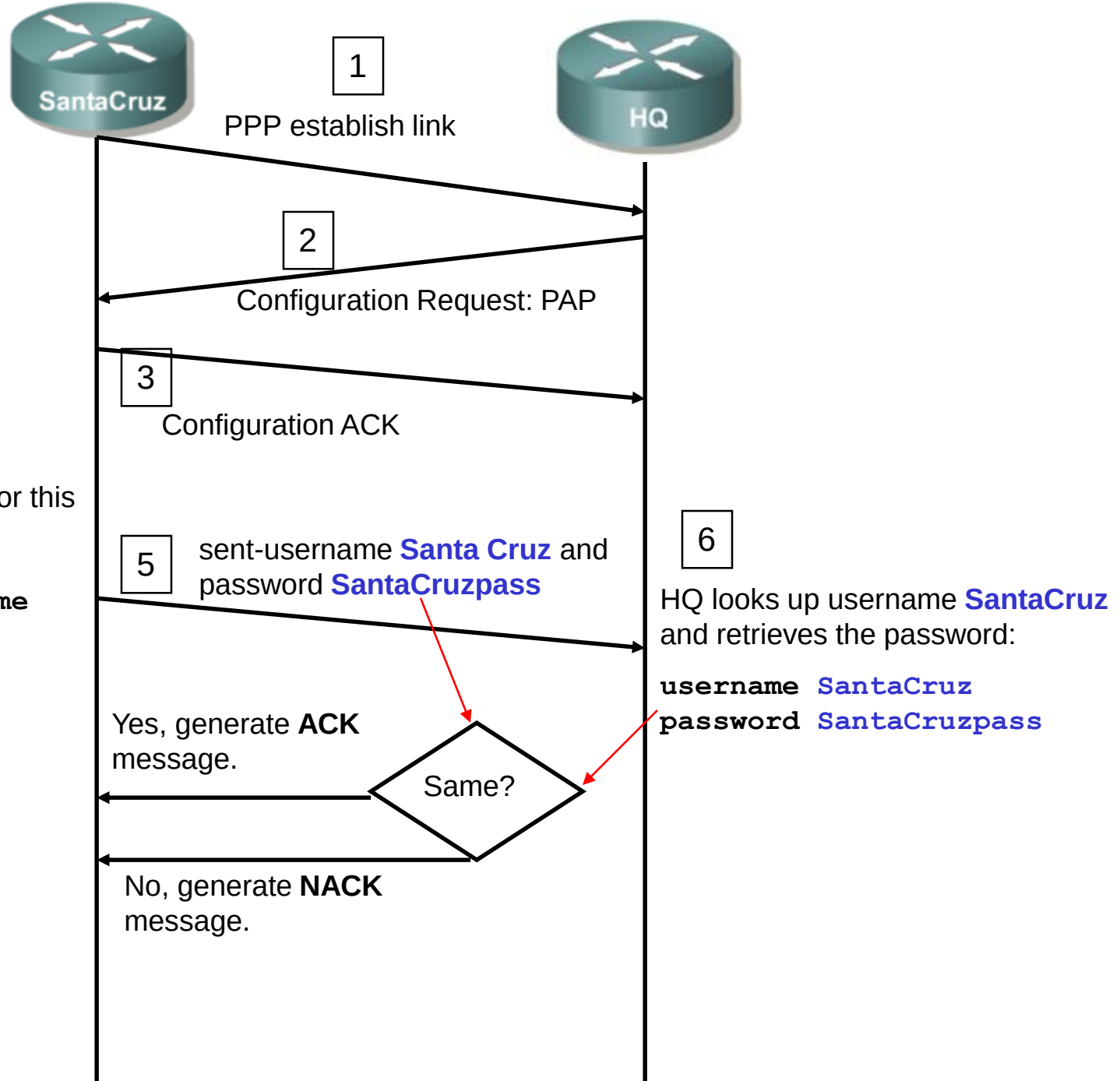
```
interface Serial0
  ip address 172.25.3.2 255.255.255.0
  encapsulation ppp
  ppp authentication pap
  ppp pap sent-username SantaCruz
    password SantaCruzpass
```

```
hostname HQ
username SantaCruz password SantaCruzpass
```

```
interface Serial0
  ip address 172.25.3.1 255.255.255.0
  encapsulation ppp
  ppp authentication pap
  ppp pap sent-username HQ
    password HQpass
```

Notes: **sent-username** and **password** must match remote **username** and **password**. Passwords are case-sensitive, but usernames are not. **Hostnames** are not involved.

PAP



Configuring CHAP



DTE
.2/S0

172.25.3.0/24
Serial

DCE
.1/S0



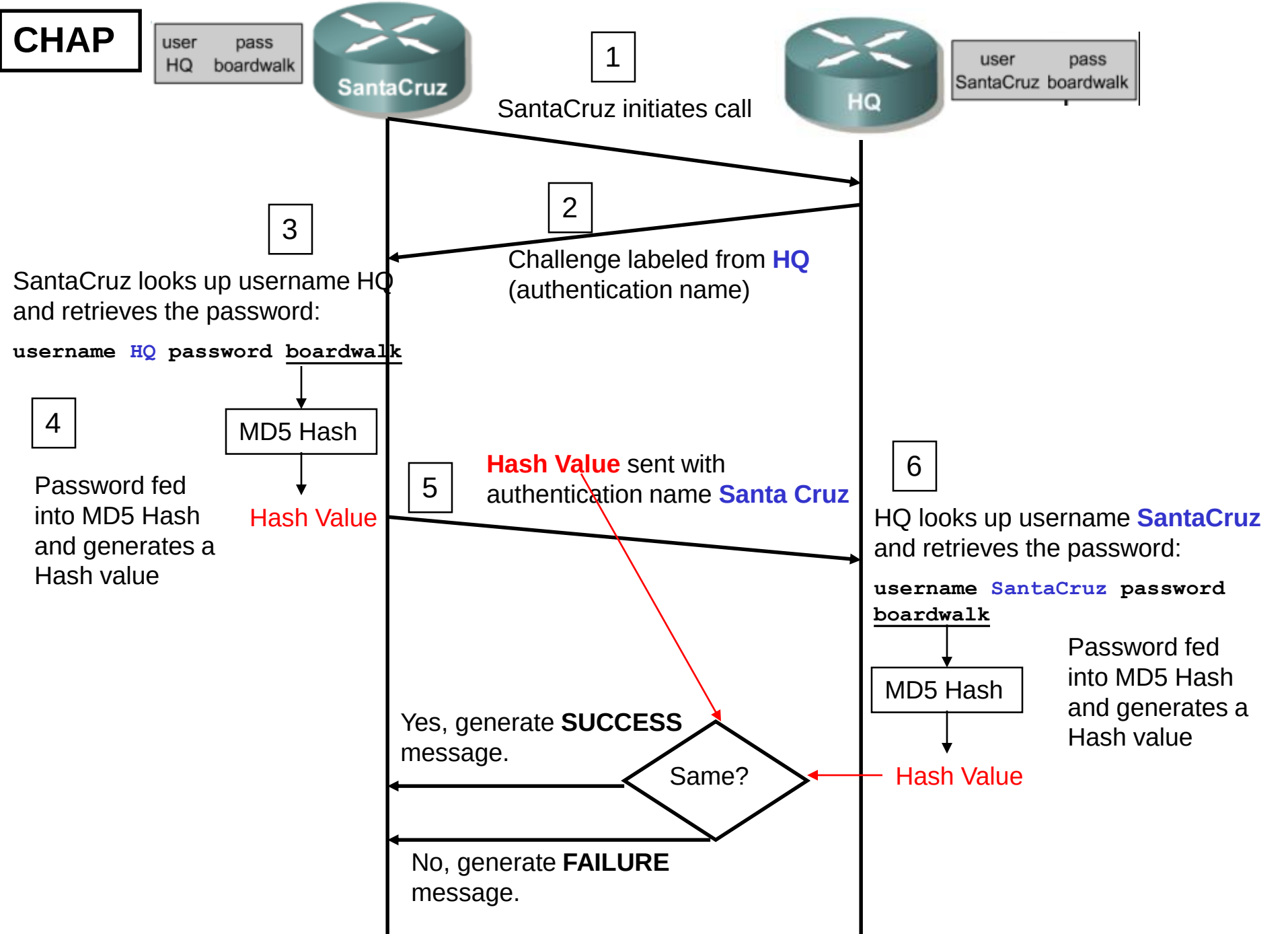
```
hostname SantaCruz
username HQ password boardwalk
ppp chap hostname SantaCruz (optional)

interface Serial0
  ip address 172.25.3.2 255.255.255.0
  encapsulation ppp
  ppp authentication chap
```

```
hostname HQ
username SantaCruz password boardwalk
ppp chap hostname HQ (optional)

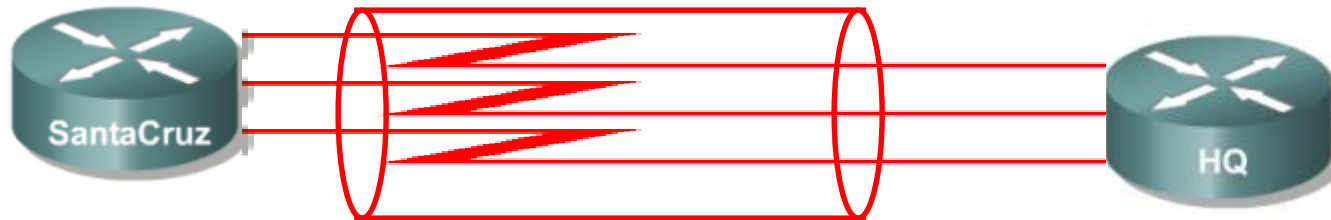
interface Serial0
  ip address 172.25.3.1 255.255.255.0
  encapsulation ppp
  ppp authentication chap
```

Notes: **Hostnames** are involved unless the **ppp chap hostname** command is used, and must match remote router's **username** command (not case-sensitive). **Passwords** are case-sensitive and must match



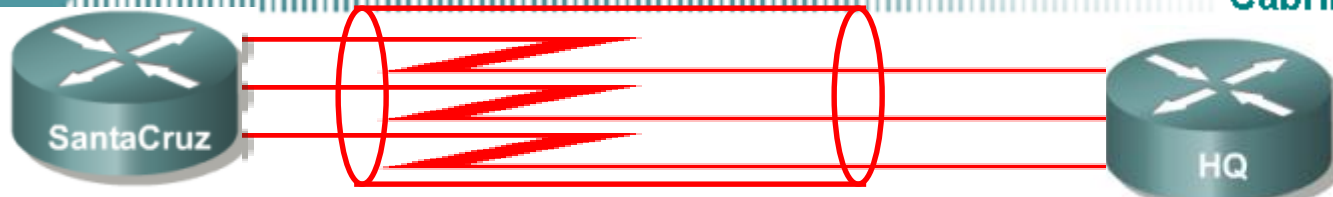
Configuring PPP Multilink (MLP)

```
Router(config)#interface serial 0/0  
Router(config-if)#encapsulation ppp  
Router(config-if)#ppp multilink
```



- In some environments, it may be necessary to bundle multiple serial links to act as single link with aggregated bandwidth.

Configuring PPP Multilink (FYI)



```
hostname SantaCruz

multilink Virtual-Template 1

interface loopback 0
 ip address 192.168.1.1 255.255.255.0

interface Virtual-Template1
 ip unnumbered loopback0
 ppp multilink
interface Serial0
 no ip address
 encapsulation ppp
 ppp multilink
interface Serial1
 no ip address
 encapsulation ppp
 ppp multilink
interface Serial2
 no ip address
 encapsulation ppp
 ppp multilink
```

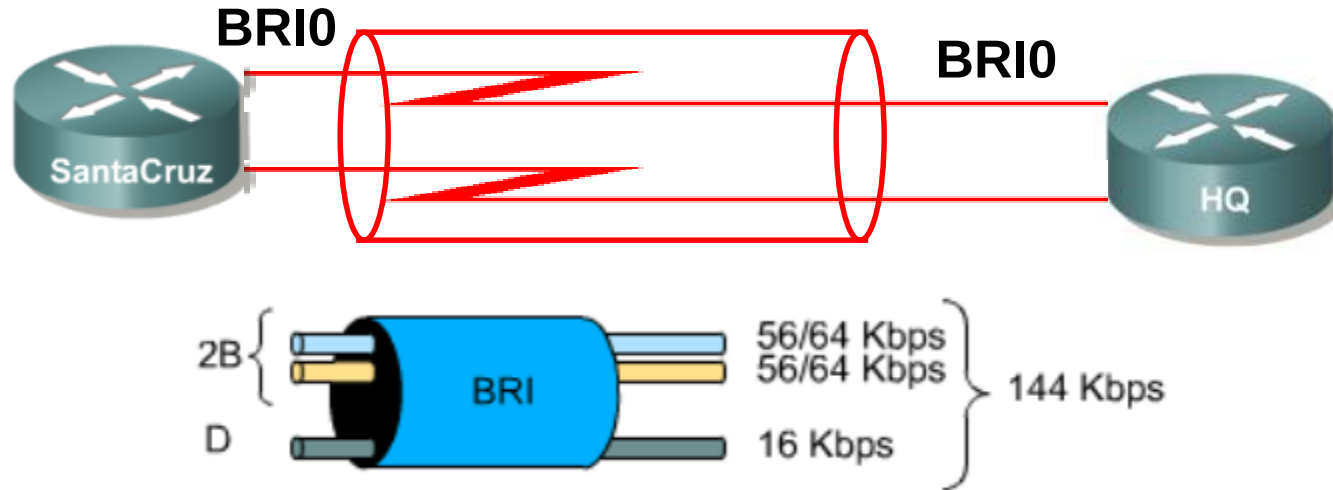
```
hostname HQ

multilink Virtual-Template 1

interface loopback 0
 ip address 192.168.1.2 255.255.255.0

interface Virtual-Template1
 ip unnumbered loopback0
 ppp multilink
interface Serial0
 no ip address
 encapsulation ppp
 ppp multilink
interface Serial1
 no ip address
 encapsulation ppp
 ppp multilink
interface Serial2
 no ip address
 encapsulation ppp
 ppp multilink
```

Configuring PPP Multilink with ISDN



- PPP Multilink is common with ISDN.
- Prior to MLP, two or more ISDN B channels could not be used in a standardized way while ensuring sequencing. MLP is most effective when used with ISDN.
- We will see how this is done when we discuss ISDN.

Configuring Compression

```
Router(config) #interface serial 0/0  
Router(config-if) #encapsulation ppp  
Router(config-if) #compress [predictor|stac|mppc]
```

- Point-to-point software compression can be configured on serial interfaces that use PPP encapsulation.
- Compression is performed in software and might significantly affect system performance.
- Compression is not recommended if most of the traffic consists of compressed files.
- To configure compression over PPP.

More Information on Compression (FYI)

Cisco supports these types of compression:

- **Predictor**-Determines whether the data is already compressed. If so, the data is just sent-no time is wasted trying to compress already compressed data.
- **Stacker**-A Lempel-Ziv (LZ)-based compression algorithm looks at the data, and sends each data type only once with information about where the type occurs within the data stream. The receiving side uses this information to reassemble the data stream.
- **MPPC**-This protocol (RFC 2118) allows Cisco routers to exchange compressed data with Microsoft clients. MPPC uses an LZ-based compression algorithm.
- **TCP header compression**-This type of compression is used to compress the TCP headers.

TCP Header Compression - RFC 1144 (FYI)

- It is supported on serial lines by using HDLC, PPP, or SLIP encapsulation.
- You must enable the compression on both ends of the connections for TCP header compression to work.
- Only TCP headers are compressed-UDP headers are not affected.
- **The data is not compressed, just the TCP header.**
- The following is the interface command used to activate TCP header compression:
 - Router(config-if) #**ip tcp header-compression**
 - The **ip tcp header-compression passive** command specifies that TCP header compression is not required, if the router receives compressed headers from a destination, then use header compression for that destination.

More Information on Compression (FYI)

Important notes on compression:

- The highest compression ratio is usually reached with highly compressible text files.
- Already compressed files such as JPEG graphics or MPEG files, or files that were compressed with software such as PKZIP or Stuffit, are only compressed 1:1, or even less.
- Trying to compress already compressed data can take longer than transferring the data without compression.
- Compressing data can cause performance degradation because it is software, not hardware compression.
- Compression can be CPU or memory intensive.
- Predictor is more memory intensive and less CPU intensive, whereas Stacker and MPPC are more CPU intensive and less memory intensive. *Memory intensive* means that an extra memory allowance is required.

Error Detection

```
Router(config)#interface serial 0/0  
Router(config-if)#encapsulation ppp  
Router(config-if)#ppp quality percentage
```

- Link Quality Monitoring (LQM) is available on all serial interfaces running PPP.
- LQM will monitor the link quality, and if the quality drops below a configured percentage, the link will be taken down.
- The percentages are calculated for both the incoming and outgoing directions.

Load Balancing

```
Router(config)#interface serial 0/0  
Router(config-if)#encapsulation ppp  
Router(config-if)#ppp multilink
```

- **Multilink PPP provides** load balancing over the router interfaces that PPP uses.
- **Packet fragmentation and sequencing**, as specified in RFC 1717, splits the load for PPP and sends fragments over parallel circuits.
- In some cases, this “**bundle**” of multilink PPP pipes functions as a single logical link, improving throughput and reducing latency between peer routers.
- Prior to MLP, two or more ISDN B channels could not be used in a standardized way while ensuring sequencing. MLP is most effective when used with ISDN.

debug ppp negotiation

```
Router#debug ppp negotiation
PPP protocol negotiation debugging is on
. . .
BR0:1 LCP: State is Open
. . .
PPP: Phase is AUTHENTICATING
. . .
BR0:1 IPCP: State is Open
. . .
```

Phase	Description
DOWN	In this phase, PPP is down. This message is seen after the link and PPP are completely brought down: *Mar 3 23:32:50.296: BR0:1 PPP: Phase is DOWN
ESTABLISHING	PPP transitions to this phase when it receives an indication that the physical layer is up and ready to be used. LCP ¹ negotiation occurs in this phase. *Mar 3 23:32:06.884: BR0:1 PPP: Phase is ESTABLISHING
AUTHENTICATING	If PPP authentication (CHAP ² or PAP ³) is desired on the link, then PPP transitions to this phase. Keep in mind that PPP authentication is optional. *Mar 3 23:32:06.952: BR0:1 PPP: Phase is AUTHENTICATING
UP	Once authentication is complete, PPP transitions to the UP phase. NCP ⁴ negotiation occurs in this phase. *Mar 3 23:42:53.412: BR0:1 PPP: Phase is UP
TERMINATING	In this phase, PPP is shutting down. *Mar 3 23:43:23.256: BR0:1 PPP: Phase is TERMINATING

- The **debug ppp negotiation** command enables you to view the PPP negotiation transactions, identify the problem or stage when the error occurs, and develop a resolution.
- During PPP negotiation, the link goes through several phases, as shown below.
- The end result is that PPP is either up or down.

debug ppp authentication

Output	Description
Se0/0 PPP: Phase is AUTHENTICATING, by both	Two way authentication
Se0/0 PAP: O AUTH-REQ id 4 len 18 from "left"	Outgoing authentication request
Se0/0 PAP: I AUTH-REQ id 1 len 18 from "right"	Incoming authentication request
Se0/0 PAP: Authenticating peer right	Authenticating incoming
Se0/0 PAP: O AUTH-ACK id 1 len 5	Outgoing acknowledgement
Se0/0 PAP: I AUTH-ACK id 4 len 5	Incoming acknowledgement

- The **debug ppp authentication** command displays the authentication exchange sequence.
- With two-way authentication configured, each router authenticates the other.
- Messages appear for both the authenticating process and the process of being authenticated.

Host Routes and PPP

Situation: When running PPP with PAP between two routers, RouterA and RouterB.

Question: When doing "show ip route" on RouterA, the routing table shows the correct network between RouterA and RouterB, BUT also shows the host ip address of RouterB as a directly connected network ("C") directly connected). Why is this happening?

Answer:

What you are seeing is normal because when the link negotiates ppp parameters, in the IPCP negotiation, they decide what IP addresses are used between them. After completion the IP address of the remote end is added in as a connected host route, which is what you are seeing in your routing table.

This is negotiated in IPCP which is the "NCP" part of PPP negotiation and happens after authentication. If you need more info, look up the RFC for PPP : 1661